UNIT 6 · STANDARD 6.EE.2A

Write Algebraic Expressions

Lesson 6-3

Name: ______________________ Date: ______________________ Class: ______________________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

x

stands for a number

In 5t + 20, t could be hours worked — if t = 3, the expression equals 35; if t = 8, it equals 60

Variable

A letter that stands for a number that is unknown or can change.

3x + 5

no equal sign

3n + 7 means 'triple a number, then add 7' — it has a variable (n), a coefficient (3), and a constant (7)

Algebraic Expression

A math phrase that has at least one letter.

4 x

coefficient = 4

In 8x, the coefficient 8 means '8 groups of x' — if x = 3, then 8x = 8 × 3 = 24

Coefficient

The number in front of a letter, like the 3 in 3x.

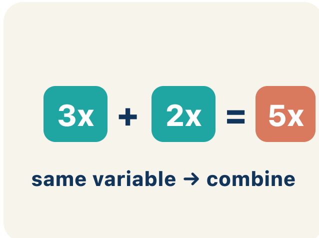
2x + 5

constant = 5 (fixed)

In 3n + 7, the 7 never changes no matter what n is — like a flat fee that stays the same

Constant

A number on its own that does not change.



4x + 2x = 6x (like terms, same variable); but 4x + 2y cannot be combined (different variables)

Like terms

Terms with the same letter, like 2x and 5x.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can write algebraic expressions from words and real-world situations.

1 Notice

You're planning a concert at the music studio.

2 Key idea

I can write algebraic expressions from words and real-world situations.

3 Words I'll use

Variable

Algebraic Expression

Coefficient **Constant**

Like terms

🤔 Think About It

- What value changes depending on how many people come?
- What stays the same no matter the number of tickets?
- How would you calculate the total if 10 tickets are sold?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which expression represents 'the product of 6 and a number n'?

A. 6n

B. 6 + n

C. n − 6

D. n ÷ 6

Step 1 'Product' means multiplication, so the product of 6 and n is 6n.

✅ **Answer:** A. 6n

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which expression represents '9 less than a number y'?

A. y − 9

B. 9 − y

C. y + 9

D. 9y

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which expression represents 'the sum of a number m and 15'?

A. m + 15

B. m − 15

C. 15m

D. m ÷ 15

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Tickets cost $12 each with a flat $50 venue fee. If t is the number of tickets, how would you write an expression for total revenue, and which part changes with the crowd?

Level 1 support

Start here: Write 12t + 50: the 12t grows with tickets, the 50 stays fixed.

💡 **Need a hint?**

Sentence starters (optional)

My expression is ____ because tickets cost $12 each.

Mi expresión es ____ porque los boletos cuestan $12 cada uno.

The 50 is a ____ because it ____.

El 50 es una ____ porque ____.

Word bank: Variable | Coefficient | Constant | Algebraic Expression

Listen for: *Students write 12t + 50, identify t as the variable number of tickets, 12 as the coefficient (price per ticket), and 50 as the constant venue fee that does not change.*

Level 2

If there were also VIP tickets at $20 each (v of them), how would your expression change? Generalize how each ticket type adds a new term.

- The new expression is ____ because each VIP ticket adds ____.
- Every new ticket type adds a term because ____.

EXPLORE

How do you turn words into an expression? Why does '12 less than a number g' become g – 12 and not 12 – g?

Level 1 support

Start here: Key words name the operation, and 'less than' reverses the order.

💡 **Need a hint?**

Sentence starters (optional)

The words '____' mean ____, so the expression is ____.

Las palabras '____' significan ____, así la expresión es ____.

It is g – 12, not 12 – g, because 'less than' means ____.

Es g – 12, no 12 – g, porque 'menos que' significa ____.

Word bank: Algebraic Expression | Variable | Coefficient | Constant

Listen for: *A strong answer matches key words to operations (sum/more than = add, less than = subtract, product = multiply) and explains '12 less than g' starts with g, giving g – 12.*

Level 2

Translate 'twice the sum of a number y and 4' and explain why it needs parentheses. How is it different from 'twice a number y, plus 4'?

- The expression is ____ because 'the sum' must be grouped, so ____.
- Without parentheses, 'twice y plus 4' would be ____, which is different because ____.

CONNECT

A studio charges a $25 setup fee plus $8 per recording hour. With h hours, what expression models the cost, and what does each part represent?

Level 1 support

Start here: Write 25 + 8h: 25 is the fixed setup, 8h grows with hours.

💡 **Need a hint?**

Sentence starters (optional)

The total cost is ___ + ___h.

El costo total es ___ + ___h.

The 25 is the ___ and 8h is the ___.

El 25 es la ___ y 8h es el ___.

Word bank: **Constant** **Variable** **Coefficient** **Algebraic Expression**

Listen for: *Students write 25 + 8h, identify 25 as the constant setup fee and 8 as the coefficient (cost per hour) multiplying the variable h.*

Level 2

Another studio charges no setup fee but $11 per hour: 11h. For how many hours is the first studio (25 + 8h) the better deal? Justify your reasoning.

- The first studio is cheaper when ___ because the higher hourly rate ___.
- Setting 25 + 8h equal to 11h, they cost the same at h = ___, so ___.

Write About the Math

The Writing Revolution

I can explain my expression using the words variable, coefficient, constant, and algebraic expression.

1. Kernel Sentence

subject + verb

Model: Algebraic Expression is a math phrase that has at least one letter.
Expresión algebraica es una frase matemática que tiene al menos una letra.

Write a kernel sentence about algebraic expression. Use a subject and a verb.
Escribe una oración base sobre expresión algebraica. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Algebraic Expression matters in math
Expresión algebraica importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Algebraic Expression matters in math because ____.** *Expresión algebraica importa en matemáticas porque ____.* ______________________________
but *pero*	**Algebraic Expression matters in math, but ____.** *Expresión algebraica importa en matemáticas, pero ____.* ______________________________
so *entonces*	**Algebraic Expression matters in math, so ____.** *Expresión algebraica importa en matemáticas, entonces ____.* ______________________________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about algebraic expression.
Di un hecho verdadero sobre algebraic expression.

Algebraic expression ____.

Question
Pregunta

Ask a question about algebraic expression.
Haz una pregunta sobre algebraic expression.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about algebraic expression.
Muestra entusiasmo sobre algebraic expression.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with algebraic expression.
Dile a un compañero qué hacer con algebraic expression.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

My expression is ____ because tickets cost $12 each.
Mi expresión es ____ porque los boletos cuestan $12 cada uno.

The 50 is a ____ because it ____.
El 50 es una ____ porque ____.

The words '____' mean ____, so the expression is ____.
Las palabras '____' significan ____, así la expresión es ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Write 'a number y divided by 4' as an expression.

A. y/4

B. 4y

C. y – 4

D. 4 – y

Show your work:

2. Which expression means '3 less than twice a number m'?

A. 2m – 3

B. 3 – 2m

C. 2(m – 3)

D. 3m – 2

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that could be modeled by the expression 4n + 10. Explain what the variable n represents, what the coefficient 4 means, and what the constant 10 represents.

Sentence starter: Situation: ____. The variable n represents ____. The coefficient 4 means ____ for each ____. The constant 10 represents ____.

Show your work:

Reflect — Exit Ticket

Which expression represents '3 more than twice a number n'?

A. 2n + 3

B. 3n + 2

C. 2(n + 3)

D. 2 + 3n

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. y – 9 *— 'Less than' means subtract from the number, so it's y – 9.*
2. **You Try (notes):** A. m + 15 *— 'Sum' means addition, so the sum of m and 15 is m + 15.*
3. **Try It 1:** A. y/4 *— 'Divided by 4' means y/4.*
4. **Try It 2:** A. 2m – 3 *— Twice a number is 2m; '3 less' subtracts 3: 2m – 3.*
5. **Exit Ticket:** A. 2n + 3 *— 'Twice a number n' is 2n. '3 more than' means + 3. So the expression is 2n + 3.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Algebraic Expression is a math phrase that has at least one letter.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).